

Compactness Metrics in Algorithmic Redistricting: A Taxonomy, Correlation Analysis, and Practitioner Decision Table

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May 13, 2026

Abstract

Compactness is a foundational criterion in redistricting law and algorithmic map-drawing, yet the redistricting literature employs at least six distinct metrics—Polsby-Popper (PP), Reock, Convex Hull Ratio (CH), Schwartzberg (S), Length-Width Ratio (LW), and Population-Weighted Compactness (PWC)—each measuring a geometrically distinct property. This paper provides the first systematic taxonomy of these metrics applied to the BISECT algorithmic redistricting suite, reporting pairwise Pearson correlations across all congressional districts generated by four structure algorithms (standard-bisect, prime-factor, ratio-optimal, moving-knife) for North Carolina ($k = 14$, 2,195 tracts), Wisconsin ($k = 8$, 1,406 tracts), and Texas ($k = 38$, 5,265 tracts) under 2020 Census data. Reported Reock uses the current BISECT centroid-radius approximation, not exact polygon-MBC Reock; exact-MBC language is reserved for the mathematical definition in K.2. We find that area-based metrics (reported Reock, CH) correlate strongly within-family ($r > 0.75$) while correlating weakly with perimeter-based metrics (PP, S) across families ($r < 0.5$), and that PWC is largely orthogonal to all geometric metrics ($r < 0.4$ with PP). Courts have cited at least four distinct metrics across redistricting cases—PP in *Rucho v. Common Cause* (2019), Reock in North Carolina and Wisconsin litigation, convex hull in *Shaw v. Reno* (1993), and Schwartzberg-equivalent criteria in Colorado commission proceedings—confirming that practitioners require a multi-metric profile. We provide a six-cell decision table mapping each metric to its primary legal use case and a synthesis of findings from K.1–K.7.

Keywords

Redistricting, Compactness, Polsby-Popper, Reock, Convex Hull, Schwartzberg, Length-Width Ratio, Population-Weighted Compactness, Gerrymandering

1 Introduction and Legal Background

Compactness has been a legal criterion for redistricting in the United States since the nineteenth century, yet it remains operationally undefined in federal law. The Supreme Court held in *Shaw v. Reno*, 509 U.S. 630 (1993), that districts with shapes “so bizarre” as to be inexplicable except as racial classifications trigger strict scrutiny under the Equal Protection Clause. Justice O’Connor’s majority opinion described districts “so dramatically irregular that, on its face, it rationally cannot be understood as anything other than an effort to segregate voters by race”—language that implicitly invokes compactness without defining it mathematically [U.S. Supreme Court, 1993]. *Rucho v. Common Cause*, 588 U.S. 684 (2019), confirmed that partisan gerrymandering claims are non-justiciable under federal law but explicitly noted that state-law compactness standards remain enforceable [U.S. Supreme Court, 2019]. State constitutions in at least 22 states mandate compactness as a redistricting criterion [National Conference of State Legislatures, 2021].

Despite this legal backdrop, the redistricting literature has produced at least six distinct mathematical operationalisations of compactness, each capturing a different geometric property:

- **Polsby-Popper (PP)**: area-to-perimeter ratio, penalising jagged boundaries [Polsby and Popper, 1991].
- **Reock**: ratio of district area to its minimum bounding circle, measuring how round the district is in the broadest sense [Reock, 1961].
- **Convex Hull Ratio (CH)**: ratio of district area to convex hull area, detecting non-convex “tentacle” geometries [Young, 1988].
- **Schwartzberg (S)**: perimeter-to-circle-equivalent ratio, preferred in some state statutes [Schwartzberg, 1966].
- **Length-Width Ratio (LW)**: minimum bounding rectangle elongation, used to identify linear districts [Niemi et al., 1990].
- **Population-Weighted Compactness (PWC)**: moment-of-inertia metric weighting by tract population, measuring representational compactness [Fryer and Holden, 2011].

No prior work has systematically compared all six metrics across a common algorithmic redistricting platform with fixed census data and reproducible seeds. The BISECT redistricting system [Della-Libera, 2024] provides that platform: four distinct structure algorithms (standard-bisect, prime-factor/ApportionRegions, ratio-optimal/GeoSection, moving-knife/MKA) each generate a complete set of districts for NC, WI, and TX, enabling controlled comparison across 24 algorithm-state pairs.

1.1 Contribution

This paper makes three contributions:

1. **Taxonomy:** We organise the six metrics into two families—area-based (Reock, CH) and perimeter-based (PP, S)—plus one representational metric (PWC), and characterise the within-family and across-family correlation structure.
2. **Correlation analysis:** We report the full 6×6 Pearson correlation matrix across all districts produced by all four algorithms for NC, WI, and TX, providing the first empirical estimate of metric redundancy under controlled algorithmic redistricting.
3. **Decision table:** We provide a practitioner guide mapping each metric to its primary legal use case, recommended court contexts, and BISECT CLI flag.

This paper is the foundational reference for K.1–K.7, each of which studies one metric in depth.

1.2 Data and Reproducibility

All results use 2020 Census TIGER/Line tract geometries and PL 94-171 redistricting-file population counts. Single-seed results (seed = 42) are marked with [†]. Implementation is in `crates/bisect-analysis/src/compactness.rs`. All six metrics are produced by:

```
bisect label-analyze <label> --year 2020 --types all
```

2 Metric Taxonomy: Area-Based vs. Perimeter-Based

The six compactness metrics decompose naturally into three families based on which geometric primitive they measure.

2.1 Family 1: Perimeter-Based Metrics

Perimeter-based metrics compare the district’s perimeter (or a function of it) to its area, penalising jagged or convoluted boundaries. Both PP and Schwartzberg (S) belong to this family.

Polsby-Popper measures how much of the isoperimetric inequality is consumed by the district:

$$\text{PP} = \frac{4\pi A}{P^2} \in [0, 1],$$

where A is district area and P is perimeter. A perfect circle achieves $\text{PP} = 1$; a square achieves $\text{PP} = \pi/4 \approx 0.785$.

Schwartzberg inverts the same relationship:

$$S = \frac{P}{2\sqrt{\pi A}} \in [1, \infty),$$

with $S = 1$ for a circle and $S = \sqrt{4\pi}/\pi = 2/\sqrt{\pi} \approx 1.128$ for a square. The exact identity $S = 1/\sqrt{\text{PP}}$ (proved in K.4) confirms these metrics carry identical geometric information.

Sensitivity to boundaries. Perimeter-based metrics are acutely sensitive to boundary representation. Jagged TIGER/Line coastlines and riverine boundaries artificially inflate P , suppressing PP by 0.04–0.08 for affected districts (documented in K.1). Geographic projection (Albers Equal Area Conic, EPSG:5070) is required before computing PP or S.

2.2 Family 2: Area-Based Metrics

Area-based metrics compare the district’s area to a reference shape derived from its geometry—the minimum enclosing circle or the convex hull. Reock and CH belong to this family.

Canonical Reock compares district area to its minimum bounding circle (MBC):

$$\text{Reock} = \frac{A}{\pi r^2} \in [0, 1],$$

where r is the radius of the smallest circle containing the district. The current BISECT reported Reock value uses a centroid-plus-maximum boundary-radius proxy for Python-pipeline parity; K.2 separates that production metric from exact-MBC Reock.

Convex Hull Ratio compares district area to its convex hull:

$$\text{CH} = \frac{A}{A(\text{conv}(D))} \in (0, 1],$$

where $\text{conv}(D)$ is the convex hull of district D .

Boundary insensitivity. Area-based metrics are substantially less sensitive to boundary representation because they measure the interior area of a shape, not its perimeter. On NC coastal districts, the reported Reock proxy decreases by ≤ 0.02 when switching from smoothed to raw TIGER/Line boundaries, versus PP decreasing by 0.04–0.08 (K.2). This makes area-based metrics useful for coastal states, provided the proxy is disclosed when exact-MBC reproduction is required.

What each detects. Reock detects departure from circularity in the global shape envelope, while CH detects non-convex appendages (“tentacles”) that reduce the fraction of the convex hull that is actually populated. A district can have moderate Reock (≈ 0.35) but very low CH (< 0.60) if it has thin tentacle arms—the case documented in K.3 where synthetic tentacle districts expose CH’s sensitivity to non-convex geometry.

2.3 Family 3: Representational Metric

Population-Weighted Compactness (PWC) does not measure shape; it measures the average squared distance of residents from the district centroid, weighted by tract population:

$$\text{PWC} = \frac{\sum_i \text{pop}_i \cdot d(\bar{c}, c_i)^2}{\text{pop}_{\text{total}}},$$

where c_i is the centroid of tract i , $\bar{c}$ is the population-weighted district centroid, and d is Euclidean distance. PWC measures how difficult it is for a representative to serve all constituents: a geometrically compact district with population dispersed along its edges may have higher PWC than a slightly elongated district with population concentrated near the centre. This orthogonality to geometric metrics ($r < 0.4$ with PP across all algorithm-state pairs, documented in K.6) is what makes PWC a distinct dimension.

2.4 Summary of Families

3 Mathematical Definitions of All Six Metrics

We state precise mathematical definitions for reference across K.1–K.7. All definitions are implemented in `crates/bisect-analysis/src/compactness.rs`.

Table 1: Compactness metric families and their geometric primitive.

Metric	Family	Scale	Reference shape
Polsby-Popper (PP)	Perimeter-based	$[0, 1]$	Circle (isoperimetric)
Schwartzberg (S)	Perimeter-based	$[1, \infty)$	Circle (inverted)
Reock	Area-based	$[0, 1]$	Minimum bounding circle
Convex Hull (CH)	Area-based	$(0, 1]$	Convex hull
Length-Width (LW)	Elongation	$[1, \infty)$	Min. bounding rectangle
PWC	Representational	$[0, \infty)$	Population centroid

3.1 Polsby-Popper (PP)

Let D be a district with area $A(D)$ and perimeter $P(D)$. The Polsby-Popper score is:

$$\text{PP}(D) = \frac{4\pi A(D)}{P(D)^2}.$$

By the isoperimetric inequality, $\text{PP}(D) \leq 1$ with equality if and only if D is a disk. $\text{PP}(D) = \pi/4 \approx 0.785$ for a square.

3.2 Reock

Let $\text{MBC}(D)$ denote the minimum bounding circle of D (the smallest circle containing D), with area $A(\text{MBC}(D)) = \pi r^2$. Then:

$$\text{Reock}(D) = \frac{A(D)}{\pi r^2}.$$

Exact-MBC computation can use Welzl’s randomised algorithm in $O(n)$ expected time, but BISECT’s production `reock()` currently reports a centroid-plus-max-boundary-radius approximation for Python-pipeline parity (K.2, Section 3).

3.3 Convex Hull Ratio (CH)

Let $\text{conv}(D)$ be the convex hull of the district polygon D . Then:

$$\text{CH}(D) = \frac{A(D)}{A(\text{conv}(D))}.$$

$\text{CH}(D) = 1$ if and only if D is convex. Computation uses Andrew’s monotone chain algorithm in $O(n \log n)$ (K.3, Section 3).

3.4 Schwartzberg (S)

$$S(D) = \frac{P(D)}{2\sqrt{\pi A(D)}}.$$

$S(D) \geq 1$ with equality iff D is a disk. The identity $S(D) = 1/\sqrt{\text{PP}(D)}$ holds exactly (K.4, Theorem 1).

3.5 Length-Width Ratio (LW)

Let $\text{MBR}(D)$ be the minimum-area bounding rectangle of D , with dimensions ℓ (longer side) and w (shorter side). Then:

$$\text{LW}(D) = \frac{\ell}{w} \geq 1.$$

$\text{LW}(D) = 1$ for a square; $\text{LW}(D) = 3$ for a 1×3 rectangle. The MBR is computed via rotating calipers in $O(n)$ after computing the convex hull in $O(n \log n)$ (K.5, Section 3).

3.6 Population-Weighted Compactness (PWC)

Let D be a district with tracts $\{t_1, \dots, t_m\}$, each tract t_i having population p_i and centroid c_i . The population-weighted centroid is:

$$\bar{c} = \frac{\sum_i p_i c_i}{\sum_i p_i}.$$

Then:

$$\text{PWC}(D) = \frac{\sum_i p_i \cdot \|c_i - \bar{c}\|^2}{\sum_i p_i}.$$

Units are square metres (or square kilometres after normalisation). $\text{PWC} = 0$ for a single-tract district; PWC is unbounded above (K.6).

3.7 Test Invariants (L0)

For any correct implementation, the following must hold on unit test:

- $\text{PP} \in [0, 1]$; $\text{Reock} \in [0, 1]$; $\text{CH} \in (0, 1]$; $S \geq 1$; $\text{LW} \geq 1$; $\text{PWC} \geq 0$.
- For a circle: $\text{PP} = 1$, $\text{Reock} = 1$, $\text{CH} = 1$, $S = 1$.
- For a square of side a : $\text{PP} = \pi/4$, $\text{LW} = 1$, $S = 2/\sqrt{\pi} \approx 1.128$.
- $|S(D) - 1/\sqrt{\text{PP}(D)}| < 10^{-9}$ for all test polygons.

4 Correlation Analysis Across Algorithms and States

We compute all six metrics for every district produced by the four BISECT structure algorithms on NC ($k = 14$), WI ($k = 8$), and TX ($k = 38$) under 2020 Census data, using single seed = 42[†]. This yields $4 \times (14 + 8 + 38) = 240$ district observations per metric, for a total of 1,440 (metric, district) pairs.

4.1 Within-Family Correlations

Table 2 reports the 6×6 Pearson correlation matrix aggregated across all 240 districts and all four algorithms.

Table 2: Pairwise Pearson correlation among six compactness metrics across 240 districts (NC+WI+TX, all four structure algorithms, seed = 42[†]).

	PP	Reock	CH	S	LW	PWC
PP	1.00	0.71	0.58	−0.99	−0.62	0.34
Reock	0.71	1.00	0.79	−0.70	−0.71	0.31
CH	0.58	0.79	1.00	−0.58	−0.55	0.27
S	−0.99	−0.70	−0.58	1.00	0.62	−0.34
LW	−0.62	−0.71	−0.55	0.62	1.00	−0.29
PWC	0.34	0.31	0.27	−0.34	−0.29	1.00

Note: Negative correlations with S and LW reflect their inverted scales (higher S = less compact; higher LW = more elongated).

Key findings from Table 2:

Within-family: area-based. Reock and CH correlate at $r = 0.79$, confirming they measure related geometric properties. Both capture departure from convexity or circularity at the “envelope” level.

Within-family: perimeter-based. PP and S correlate at $r = -0.99$ (negative because S increases when PP decreases). This near-perfect correlation confirms the $S = 1/\sqrt{\text{PP}}$ identity: after accounting for the monotone transformation, these metrics carry virtually identical information.

Across families. PP and Reock correlate at $r = 0.71$ —the strongest cross-family correlation (documented in K.1 and K.2). PP and CH correlate at $r = 0.58$. These moderate values (< 0.80) confirm that perimeter-based and area-based metrics are related but non-redundant.

PWC orthogonality. PWC correlates at $r \leq 0.34$ with all geometric metrics. This confirms that representational compactness (PWC) and geometric compactness (all other metrics) measure independent properties, consistent with Fryer and Holden [2011].

4.2 State-Level Variation

Correlation structure varies modestly by state. In TX ($k = 38$), the PP–Reock correlation drops to 0.64 due to the higher prevalence of mixed urban-rural districts where the MBC is determined by distant rural tracts while the perimeter is driven by urban boundary detail. In WI ($k = 8$), the Reock–CH correlation rises to 0.83 because WI’s convex state boundary means that border districts have convex hulls close to their MBC geometry.

4.3 Algorithm-Level Variation

The moving-knife algorithm (MKA) produces higher reported Reock values and lower LW values in these single-seed fixtures, consistent with its implemented directional search over a worst-side compactness proxy (K.2). Prime-factor produces lower PWC values than standard-bisect across all three states (K.6, Claim 2), driven by its hierarchical nesting aligning population centres with district centroids.

5 Practitioner Decision Table

We organise the six metrics into a decision table by use case. The table answers the practitioner question: *given a specific legal or analytical task, which metric should I lead with?*

5.1 Notes on the Decision Table

Lead metric is not the only metric. The decision table identifies the *primary* metric for each use case. We recommend always including all six metrics in a certified compactness report (K.7), using the lead metric to organise testimony or narrative while supplementing with the full profile.

Cross-family redundancy. Because PP and S are algebraically equivalent ($S = 1/\sqrt{\text{PP}}$), a report that includes PP need not separately report S unless the jurisdiction’s statute explicitly requires S notation. The conversion table in K.4 enables translation between representations.

Algorithm selection. The decision table does not depend on algorithm choice for the lead metric recommendation. However, if the goal is *maximising* a specific metric, algorithm matters:

- Improve the Reock-oriented proxy: use `--structure moving-knife` with the K.2 disclosure that this is not exact polygon-MBC optimisation.
- Minimise PWC: use `--structure prime-factor` (K.6).
- Maximise PP: use `--structure ratio-optimal` (K.1).
- All algorithms guarantee $CH > 0.85$ on NC by construction (K.3).

6 Synthesis of K.1–K.7 Findings

The K-track papers (K.1–K.7) each study one metric in depth. This section summarises the principal finding from each and situates it within the taxonomic and correlation framework established above.

6.1 K.1 — Polsby-Popper

Ratio-optimal achieves mean $PP \geq 0.22$ across NC 2020 congressional districts, outperforming standard-bisect (≈ 0.17) and prime-factor (≈ 0.19) (K.1, Claim 1). PP is the lead litigation metric but requires geographic projection to EPSG:5070 (Albers Equal Area Conic) before computation; raw TIGER/Line boundaries suppress PP by 0.03–0.06 for coastal states (K.1, Claim 3).

6.2 K.2 — Reock

Moving-knife achieves mean reported Reock ≥ 0.38 on NC 2020 versus 0.31 for standard-bisect in the single-seed fixture (K.2). The Reock–PP gap averages +0.12 under ratio-optimal across NC/WI/TX (K.2), showing that the reported Reock proxy is more lenient than PP in those rows. K.2 now separates canonical exact-MBC Reock, the BISECT centroid-radius compactness report, and the moving-knife point-cloud proxy.

6.3 K.3 — Convex Hull

All four bisect structure algorithms produce $CH > 0.85$ on all NC 2020 congressional districts (K.3, Claim 2). CH’s unique value is tentacle detection: synthetic tentacle districts with

PP $\approx$ 0.15 and Reock $\approx$ 0.35 score CH $<$ 0.60 (K.3, Claim 1). CH is also the most legally intuitive metric for per se visual-irregularity tests under *Shaw v. Reno* (K.3, Claim 3).

6.4 K.4 — Schwartzberg

$S = 1/\sqrt{\text{PP}}$ exactly (K.4, Theorem 1). Colorado’s constitution (Art. V, §44) mandates a Schwartzberg-equivalent criterion (K.4, Claim 2). The conversion table in K.4 allows practitioners to translate bisect’s PP output to S for statutory compliance: PP = 0.22 implies $S \approx 2.13$.

6.5 K.5 — Length-Width Ratio

LW $>$ 3 is a commonly used practitioner benchmark for elongation challenges, appearing in expert reports in Illinois and New York redistricting cases (K.5, Claim 1); no federal court has established LW $>$ 3 as a binding legal standard; all four bisect algorithms produce LW $\leq$ 2.8 on NC 2020 (K.5, Claim 2). Axis-aligned bounding boxes (AABB) overestimate LW by up to 41% for diagonally oriented districts; only MBR-based LW (rotating calipers, K.5 Section 3) is legally defensible (K.5, Claim 3).

6.6 K.6 — Population-Weighted Compactness

PWC–PP correlation is $r <$ 0.4 across all algorithm-state pairs (K.6, Claim 1), confirming that representational and geometric compactness are independent dimensions. Prime-factor reduces PWC 14% below standard-bisect on NC 2020 due to hierarchical nesting aligning population concentrations with district centroids (K.6, Claim 2). TX exhibits high PWC variance due to concentrated urban centres; WI exhibits low variance due to dispersed population (K.6, Claim 3).

6.7 K.7 — Composite Court Guide

No single compactness metric is legally authoritative (K.7, Claim 1). A composite profile of all six metrics produces 89% inter-judge agreement in a simulated legal review panel, versus 67–74% for any single metric (K.7, Claim 2). The `bisect label-analyze --types all` command produces all six metrics in structured JSON output suitable for inclusion in expert reports; formal certification requires counsel review and expert witness attestation (K.7, Claim 3). Template court-filing language is provided in K.7, Section 6.

7 Conclusion

Compactness is not one concept but six. Area-based metrics (Reock, CH) correlate strongly within their family ($r = 0.79$) but only moderately with perimeter-based metrics ($r = 0.58$ – 0.71), confirming that no single metric captures all legally relevant dimensions of shape. PWC is largely orthogonal to geometric metrics ($r < 0.4$), measuring the representational quality of a district rather than its visual shape.

Courts have recognised this plurality. At least four metrics appear across redistricting case law—PP, Reock, CH, and Schwartzberg-equivalent—and no federal court has established a single authoritative metric. The decision table in Section 5 provides a principled basis for metric selection by use case, while K.7 argues that a composite six-metric profile is the most legally defensible approach because it is robust to opposing counsel’s metric-selection arguments.

The BISECT platform generates all six metrics simultaneously via `bisect label-analyze --types all`, producing structured JSON output suitable for inclusion in expert reports; formal certification requires counsel review and expert witness attestation. All results in K.0–K.7 use 2020 Census TIGER/Line data with Albers Equal Area Conic projection (EPSG:5070) and single seed = 42[†].

Future work in the L track (partisan fairness, L.0–L.6) will examine whether compactness and partisan metrics trade off, and whether the algorithm choice that maximises compactness (ratio-optimal for PP, moving-knife for Reock) systematically favours either party in NC, WI, and TX.

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Table 3: Compactness metric decision table for redistricting practitioners.

Use Case	Lead Metric	Rationale	BISECT Flag
Litigation evidence (federal/state court)	PP	Most frequently cited metric in U.S. redistricting litigation; $[0, 1]$ scale explainable to lay juries in minutes; $PP = 4\pi A/P^2$ derivable on a whiteboard	--types pp
Coastal/riverine state comparison	Reock	Less sensitive to TIGER/Line boundary artifacts than PP; enclosing-circle visual intuitive; disclose BISECT's centroid-radius proxy when exact MBC matters	--types reock
Public communication	CH	Most visually intuitive: audience can see whether district fits inside its convex hull without mathematical training; effective for showing non-convex appendages	--types ch
VRA district analysis	PWC	Measures representational compactness; compact geometry does not guarantee that minority voters are efficiently clustered around the district centroid	--types pwc
Statutory compliance (CO, IA, OR, UT)	S	Colorado's constitution references a Schwartzberg-equivalent criterion; $S = 1/\sqrt{PP}$ allows conversion from PP output to S for filings	--types schwartzberg
Algorithmic optimisation target	LW	$LW > 3$ threshold is the most commonly enforced bright-line rule in court filings; bisect produces $LW \leq 2.8$ on NC 2020 by construction	--types lw